

Assignment 4: PII, HIPAA, and Privacy

Ethics and Compliance of AI

1. **Employing the Safe Harbor Method:** You are given a file of patient data including PII, demographic information, and diagnosis codes from visits to a clinic. This file is *random_patient_data.tsv*, a tab delimited file. You want to pass this file to a researcher who will study it for relationships between demographics and illness. It is important that you preserve the ability to keep the relationships between the demographic information and the diagnosis codes intact. You also want to ensure that each row can be uniquely identified. However, you need to redact the dataset so no information that can identify the individual in each row is passed to the researcher. You decide to use the Safe Harbor method to redact the file.
 - a. Which fields in the dataset have to be redacted? (9 pts)
 - b. Describe how you will redact each field. (10 pts)

2. **Re-identifying Data:** It's possible to use different combinations of data from different files to determine the identity of otherwise unknown individuals. In this case, you will play a logic puzzle to properly match each individual to his or her disease.
 - a. The Setup: Five individuals – Alex, Ben, Chloe, David, and Emily – each have a unique age (30, 35, 40, 45, 50) and a unique disease (Asthma, Diabetes, Gout, Hypertension, Migraines). Your task is to match each person to their correct age and disease.
 - b. The Clues:
 - i. Ben is exactly 10 years older than the person with Gout.
 - ii. The person with Diabetes is younger than Chloe, but older than the person who is 30.
 - iii. Emily is 45, but she does not have Asthma or Migraines.
 - iv. Alex is not 50 and does not have Hypertension.
 - v. David is older than the person with Asthma, but younger than the person with Migraines.
 - vi. The 30-year-old does not have Hypertension.
 - vii. Chloe is not 35 and does not have Diabetes.
 - c. Match each individual to his or her age and disease. (Each correct individual/age/disease combination is worth 5 points for a total of 25 points)

3. **Re-identification with Probability:** Usually it isn't possible to explicitly re-identify an individual like it was in the logic puzzle. Instead, the probability that an individual has an identify or feature is determined. In this problem, you will determine the probability that customers have had COVID based upon data that doesn't directly state their COVID history.
- a. There are three individuals and three zipcodes as described in the tables below:

Zip Code	Percent of Population That Has Had COVID	
	Residents Under 50	Residents 50 and Older
97130	35%	25%
97141	25%	15%
97150	15%	10%

Customer	Probability Of Living In Zip Code			Age
	97130	97141	97150	
Amy	0.62	0.28	0.20	27
Bill	0.15	0.55	0.30	52
Charlotte	0.20	0.10	0.70	71

Calculate the probability that each customer has had COVID. Show your work – if you don't show your work, you receive no points. Hint: you will use conditional probabilities to determine the outcome. (21 points total)

- b. In (3a), you were given the probability that each customer lived in a given zipcode. Suppose that you don't know these probabilities but instead must derive them from other data. One way to do that would be with location data. You work for a telecommunications company, and you have a file containing customers and the location of the origination of their phone calls. You assume that the probability of their home location is equal to the fraction of calls originating from that location. You know the probability of COVID by age and zipcode from (3a). You also know the customers' ages from doing credit checks. Write a python program to calculate the probability that each customer has had COVID. The phone call data is given to you in the file *customer_phone_calls.tsv*. You will fill in the missing code in the code skeleton *aai4323_hw4_lastname.py*. When you submit your assignment, make sure to change the name of your file by replacing "lastname" with your actual last name (35 points total).

NOTES ON THE CODING ASSIGNMENT:

- Ensure you have conda and an integrated development environment (I recommend Visual Studio Code) installed on your computer
- Copy all the files into a single directory
- Create a conda environment using the YAML file: `conda env create -f aai4323_conda_all.yml`
- Go to the directory where you stored the homework files
- Activate the conda environment: `conda activate aai4323`
- Change the name of the file `aai4323_hw4_lastname.py` by replacing “lastname” with your last name.
- Open your IDE and edit the file
- Submit your completed script, along with a PDF including your answers to questions 1-3a to Canvas.